A MAP OF PERIODIC COMPLEXES WHOSE KERNEL AND COKERNEL HAVE FINITE LENGTH DOES NOT CHANGE THE MULTIPLICITY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Additivity of the multiplicity for a short exact sequence of periodic complexes. Same chapter chow.tex, label lemma-additivity-periodic-length (invoked at proof line 291). Applied to the sequence formed by the kernel and cokernel of f, it expresses $e_R(M, \varphi, \psi)$ as the sum of $e_R(M', \varphi', \psi')$ and the multiplicities of the two end terms.
- Vanishing of the multiplicity for a periodic complex with finite length cohomology. Same chapter chow.tex, label lemma-finite-periodic-length (invoked at proof line 294). It shows that the kernel and the cokernel, being of finite length and with finite length cohomology, contribute zero.

Source reference. Original lemma 0EA7: sources/stacks-chow/pinned-original-source.tex, source lines 209–209. Navigation pointer for \ref{lemma-additivity-periodic-length} used inside the counted span (line 291); the argument that consumes it is retained in the span.

Source reference. Original lemma 0EA8: sources/stacks-chow/pinned-original-source.tex, source lines 258–258. Navigation pointer for \ref{lemma-finite-periodic-length} used inside the counted span (line 294); the argument that consumes it is retained in the span.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let R be a ring. Let $f : (M, \varphi, \psi) \to (M', \varphi', \psi')$ be a map of $(2,1)$-periodic complexes whose cohomology modules have finite length. If $\mathrm{Ker}(f)$ and $\mathrm{Coker}(f)$ have finite length, then $e_R(M, \varphi, \psi) = e_R(M', \varphi', \psi')$.*

Proof. Apply the additivity of Lemma 0EA7 and observe that $(\mathrm{Ker}(f), \varphi, \psi)$ and $(\mathrm{Coker}(f), \varphi', \psi')$ have vanishing multiplicity by Lemma 0EA8. □